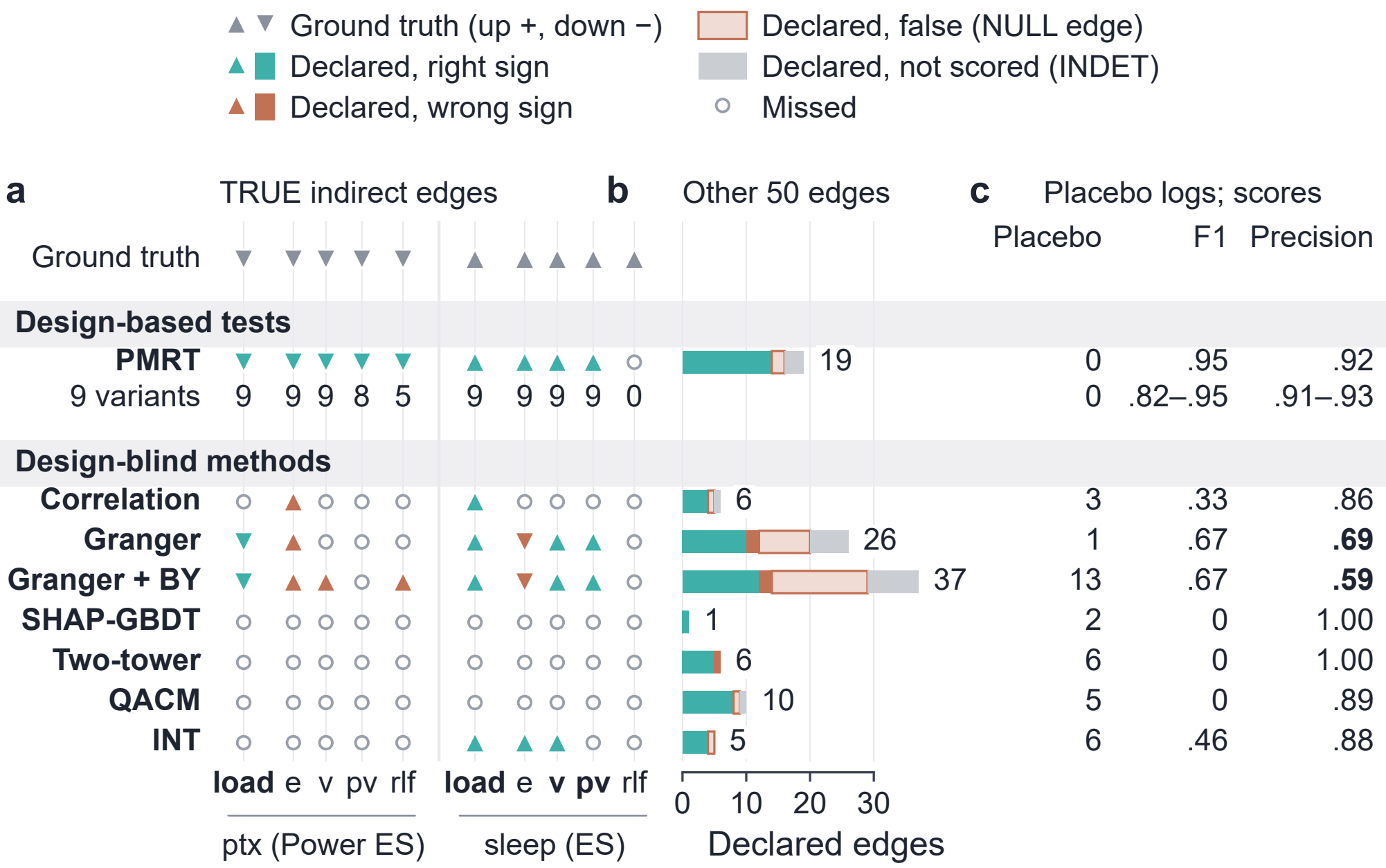

Ground truth (up +, down −)
Declared, right sign
Declared, wrong sign
Declared, false (NULL edge)
Declared, not scored (INDET)
Missed
a
TRUE indirect edges
b
Other 50 edges
c
Placebo logs; scores
Placebo
F1
Precision
Ground truth
Design-based tests
PMRT
9 variants
9 9 9 8 5
9 9 9 9 0
19
0
.95
.92
0
.82–.95
.91–.93
Design-blind methods
Correlation
6
3
.33
.86
Granger
26
1
.67
.69
Granger + BY
37
13
.67
.59
SHAP-GBDT
1
2
0
1.00
Two-tower
6
6
0
1.00
QACM
10
5
0
.89
INT
5
6
.46
.88
load e v pv rlf
load e v pv rlf
ptx (Power ES)
sleep (ES)
0 10 20 30
Declared edges